Properties of MLE: Uncertainty in Parameters

INFO 521 · Module 3 · Lecture B — Properties of the MLE

Greg Chism
School of Information · University of Arizona

July 2026

Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

Learning outcomes

By the end of this module you should be able to:

1. **Show the MLE solution is unique** — that the log-likelihood has a single maximum (a negative-definite Hessian).
2. **State what unbiasedness of $\hat{\mathbf{w}}$ does and does not mean** — and the common claims it is *not*.
3. **Quantify parameter uncertainty** — via $\mathrm{Cov}[\hat{\mathbf{w}}]$ and the Fisher information.

Recap: likelihood and the MLE

The linear-Gaussian likelihood:

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) \;=\; \prod_{n=1}^{N} \mathcal{N}\left(y_n \,\middle|\, \mathbf{w}^\top \mathbf{x}_n,\ \sigma^2\right)$$

Its maximizers — the MLE (derived in the likelihood lecture of this module):

$$\hat{\mathbf{w}} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}, \qquad \hat{\sigma}^2 = \tfrac{1}{N}\left(\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\hat{\mathbf{w}}\right)$$

Unique, unbiased, and how uncertain.

Property 1 · Uniqueness

The log-likelihood — we write it $\ell(\mathbf{w}) = \log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2)$, reserving $\mathcal{L}$ for losses:

$$\ell(\mathbf{w}) = -\tfrac{N}{2} \log 2\pi \;-\; N \log \sigma \;-\; \tfrac{1}{2\sigma^2} \sum_{n=1}^{N} \left(y_n - \mathbf{w}^\top \mathbf{x}_n\right)^2$$

Gradient — vanishes at the normal equations:

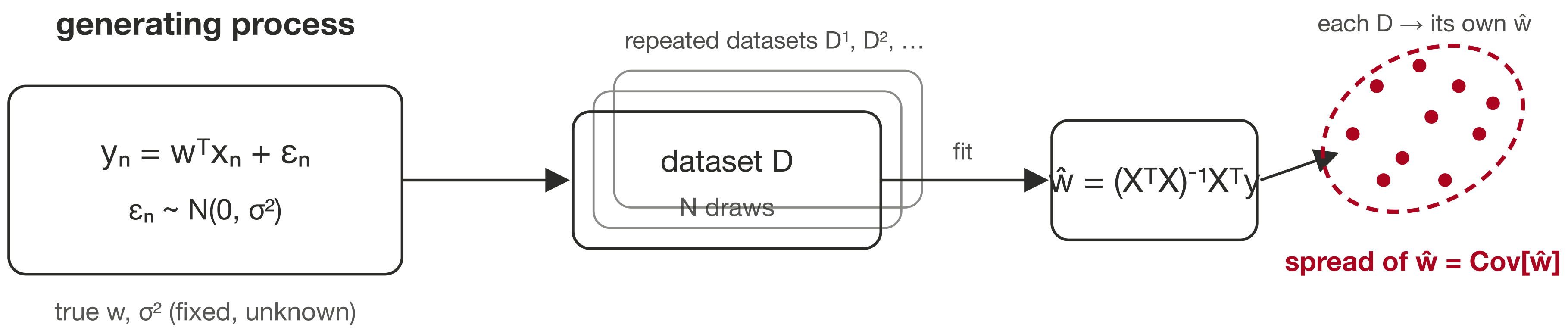
$$\frac{\partial \ell}{\partial \mathbf{w}} = \tfrac{1}{\sigma^2} \left(\mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \mathbf{w}\right)$$

Hessian — constant and **negative definite** ($\mathbf{v}^\top \mathbf{H} \mathbf{v} < 0 \;\; \forall\, \mathbf{v} \neq \mathbf{0}$):

$$\mathbf{H} \;=\; \frac{\partial^2 \ell}{\partial \mathbf{w}\, \partial \mathbf{w}^\top} \;=\; -\tfrac{1}{\sigma^2}\, \mathbf{X}^\top \mathbf{X} \quad \Longrightarrow \quad \text{a single, unique maximum.}$$

The same $\mathbf{X}^\top \mathbf{X}$ full-column-rank condition from Module 1 — now reappearing as a negative-definite Hessian.

The generative picture



Expectation — a reminder

For a random vector $\mathbf{x}$ with density $p(\mathbf{x})$:

$$\mathbb{E}_{p(\mathbf{x})}[f(\mathbf{x})] = \int f(\mathbf{x})\, p(\mathbf{x})\, d\mathbf{x}$$

Mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$ **Covariance** $\mathrm{Cov}[\mathbf{x}] = \mathbb{E}\left[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top\right]$

Property 2 · Unbiasedness

Take the expectation of $\hat{\mathbf{w}}$ over datasets from the generating process (only $\mathbf{y}$ is random; $\mathbf{X}$ is fixed):

$$\begin{aligned}\mathbb{E}[\hat{\mathbf{w}}] &= \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbb{E}[\mathbf{y}] \\ &= \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{X} \mathbf{w} \\ &= \mathbf{w} \qquad \Longrightarrow \qquad \hat{\mathbf{w}} \text{ is unbiased.}\end{aligned}$$

What “unbiased” means — and doesn’t

Unbiased means: across repeated size-N samples from the generating process, the collection of estimates $\hat{\mathbf{w}}$ is **centered** on the true $\mathbf{w}$.

It is **not** the same as:

- **“more data $\Rightarrow$ better estimates”** — true here, but that is a *separate* sample-size effect, not what unbiasedness asserts;
- **UMVU** (uniformly minimum-variance unbiased) — a strictly stronger claim about being *closest*, which unbiasedness alone does not give;
- a guarantee of being best for *your* dataset — a **biased** estimator can land closer to $\mathbf{w}$ from *less* data.

Property 3 · Covariance

The punchline first — the parameter covariance is

$$\boxed{\mathrm{Cov}[\hat{\mathbf{w}}] \;=\; \sigma^2 \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}}$$

Derived in full in the appendix → appendix-covariance.html.

Fisher information

The (expected) Fisher information is the curvature of the log-likelihood:

$$\mathcal{I} \;=\; \mathbb{E}\left[-\,\frac{\partial^2 \log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2)}{\partial \mathbf{w}\, \partial \mathbf{w}^\top}\right] \;=\; \tfrac{1}{\sigma^2}\, \mathbf{X}^\top \mathbf{X} \qquad \Longrightarrow \qquad \mathrm{Cov}[\hat{\mathbf{w}}] \;=\; \mathcal{I}^{-1}$$

- **Diagonal** $\mathcal{I}_{jj}$ — information about each parameter on its own;
- **Off-diagonal** $\mathcal{I}_{jk}$ — information *shared* between a pair (their estimates trade off).

Parameter uncertainty on NHANES

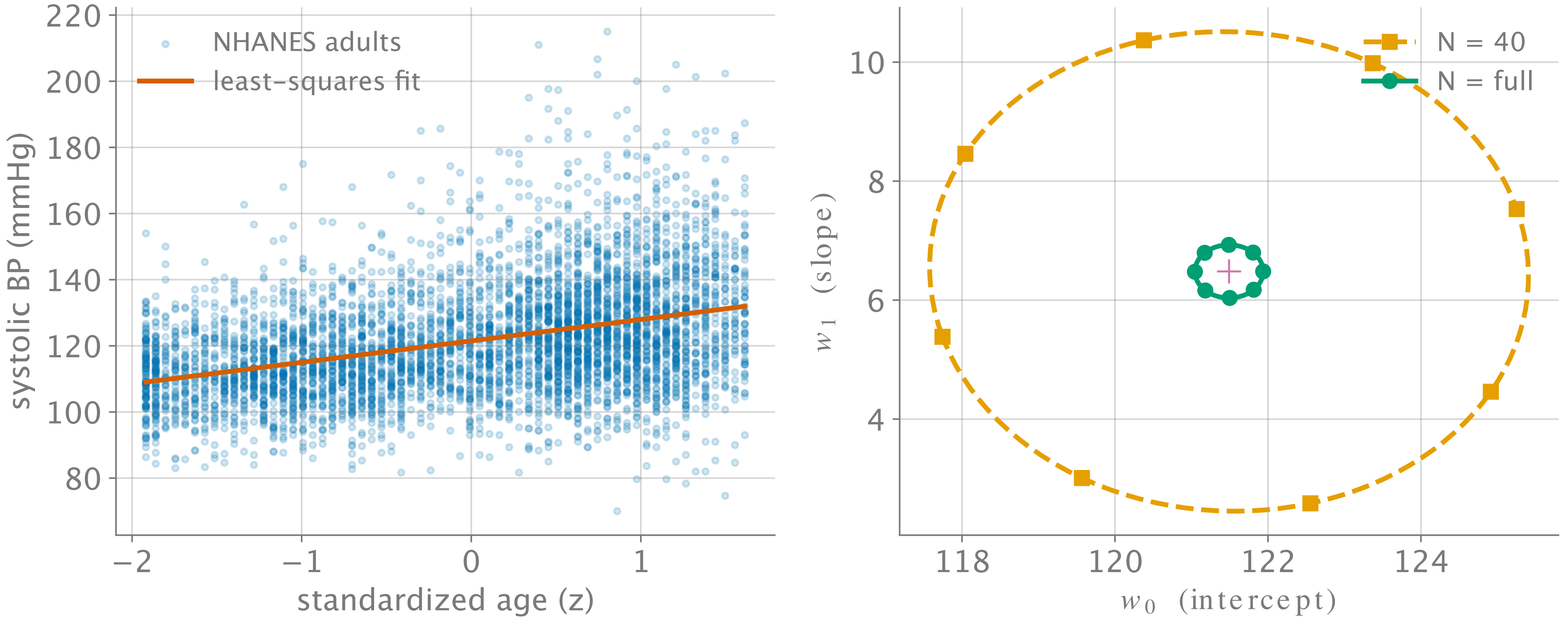


Figure 1: $\mathrm{Cov}[\hat{w}] = \hat{\sigma}^2(X^\top X)^{-1}$ shrinks as N grows: N = 40 (orange, dashed) vs. full N (green, solid).

Through-line

Uniqueness (negative-definite Hessian) $\rightarrow$ Fisher information $\mathcal{I} = \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}$ $\rightarrow$ (Module 5) the Laplace approximation's posterior covariance $\mathbf{H}^{-1} = \mathcal{I}^{-1}$.

The curvature of the log-likelihood is one object doing three jobs, three modules apart — **same $\mathbf{X}^\top\mathbf{X}$, same inverse.**

Recap + checkpoint preview

Three properties of the linear-Gaussian MLE:

1. **Unique** — $\mathbf{H} = -\frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}$ is negative definite.
2. **Unbiased** — $\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}$.
3. **Uncertain, quantifiably** — $\mathrm{Cov}[\hat{\mathbf{w}}] = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} = \mathcal{I}^{-1}$.

The **P1-M2 closed-book checkpoint** gates on reproducing the **uniqueness** and **unbiasedness** arguments from scratch.